

Training Results — Milestone 1 Classification Baseline

1. Setup Recap

5 CelebA identities selected for balanced image counts (30 each) and visual diversity across gender, hair color, and eyewear/headwear:

Identity ID	# Images	Dominant Attributes
6439	30	Blond Hair
5640	30	Male, Gray Hair, Eyeglasses, Bald
6195	30	Male, Black Hair, Mustache
5485	30	Male, Eyeglasses, Wearing Hat
7087	30	Black Hair, Eyeglasses

Split: stratified 70/15/15 per identity → 110 train / 20 val / 20 test images

Preprocessing: 128x128 resize, ImageNet normalization, light train-only augmentation (horizontal flip, light color jitter)

Models compared: (1) Custom CNN from scratch, (2) ResNet18 fine-tuned (ImageNet-pretrained)

2. Model Comparison Table

Model	Test Acc.	Val Acc.	Params	Size (MB)	Train Time (s)
ResNet18 (fine-tuned)	0.95	1.00	11,179,077	42.64	456.3
Custom CNN (from scratch)	0.85	0.85	1,207,717	4.61	1077.0

3. Per-Class Performance

ResNet18 (fine-tuned) — best model

Identity	Precision	Recall	F1-score	Support
5485	1.00	0.75	0.857	4
5640	1.00	1.00	1.000	4
6195	0.80	1.00	0.889	4
6439	1.00	1.00	1.000	4
7087	1.00	1.00	1.000	4
accuracy			0.95	20
macro avg	0.96	0.95	0.949	20
weighted avg	0.96	0.95	0.949	20

Only one image (identity 5485) was misclassified — as identity 6195, the only class with imperfect precision (0.80).

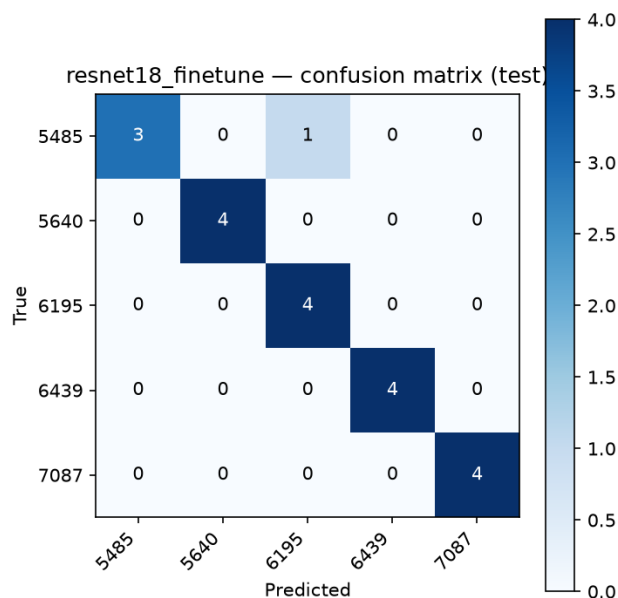


Figure 1: ResNet18 fine-tuned confusion matrix (test set)

Custom CNN — for comparison

Identity	Precision	Recall	F1-score	Support
5485	0.75	0.75	0.750	4
5640	1.00	1.00	1.000	4
6195	0.60	0.75	0.667	4
6439	1.00	1.00	1.000	4
7087	1.00	0.75	0.857	4
accuracy			0.85	20
macro avg	0.87	0.85	0.855	20
weighted avg	0.87	0.85	0.855	20

The custom CNN made 3 misclassifications, all clustered around identity 6195 — suggesting its learned features for that identity overlap with others, unlike ResNet18's more cleanly separated feature space.

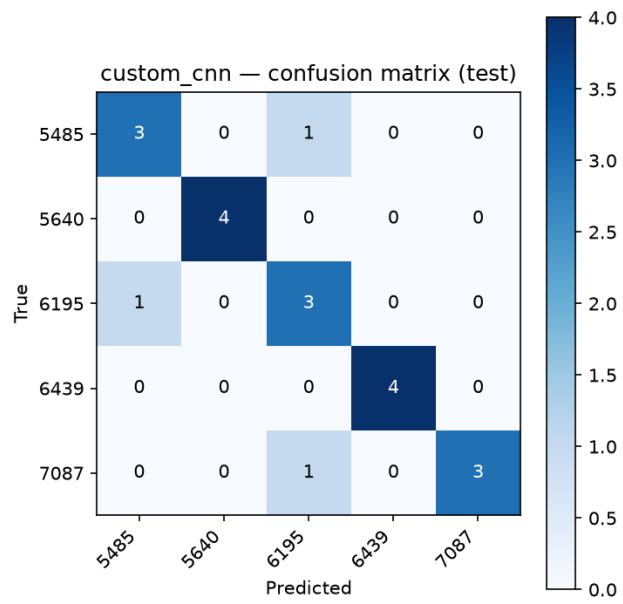


Figure 2: Custom CNN confusion matrix (test set)

4. Training Curves

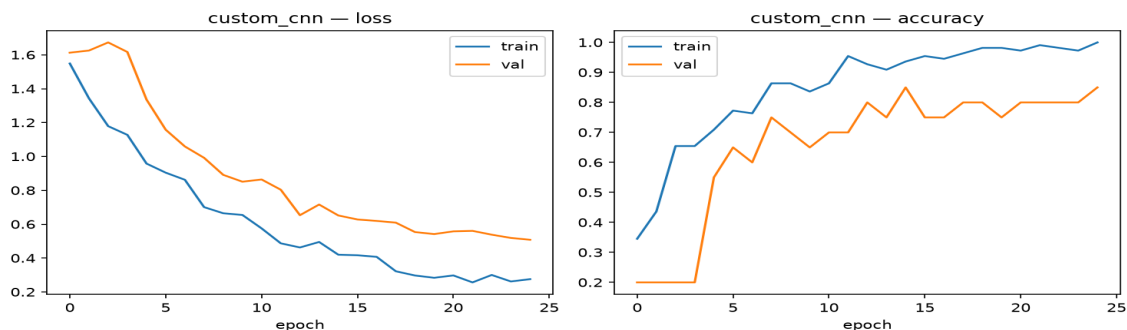


Figure 3: Custom CNN training/validation loss and accuracy (25 epochs, tuned hyperparameters)

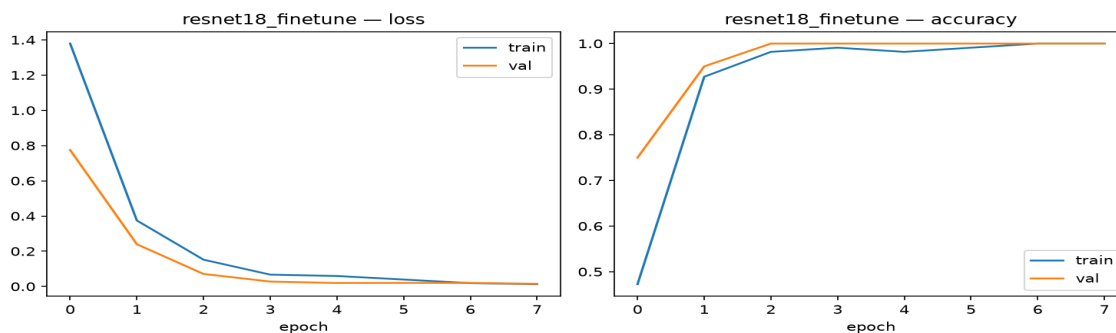


Figure 4: ResNet18 fine-tuned training/validation loss and accuracy

The ResNet18 fine-tune shows both training and validation loss dropping sharply and in lockstep, reaching 100% validation accuracy by epoch 3. The custom CNN's curves show healthy, gradual learning once augmentation was lightened and the learning rate lowered: training and validation loss both decrease steadily together, and validation accuracy climbs from 20% to 85% over 25 epochs — a legitimate learning curve, in contrast to an earlier training run where validation accuracy stayed flat at exactly 20% every epoch while training accuracy climbed, indicating the model had collapsed to predicting a single class.

5. Best-Model Justification

The **ResNet18 fine-tuned model** is the model we carry forward into Milestone 2/3, achieving 95% test accuracy versus the custom CNN's 85%. Both models now learn genuine, generalizing features — a meaningfully different result than our first training pass, where the custom CNN's original hyperparameters (heavier color jitter + random rotation augmentation, learning rate 1e-3, dropout 0.4, batch size 32) caused it to collapse to predicting a single class for every input (20% accuracy, equivalent to random guessing among 5 classes). Lightening the augmentation, lowering the learning rate to 3e-4, reducing dropout to 0.3, and training with a smaller batch size (16) for longer (up to 40 epochs, patience 10) let the same architecture actually learn: 85% test accuracy, with training and validation loss decreasing together rather than diverging.

Even with this fix, ResNet18 still outperforms the custom CNN by 10 percentage points, which we attribute to its ImageNet-pretrained features already encoding general-purpose visual structure that fine-tuning only needs to adapt to our 5 identities, rather than learning vision from zero on ~22 training images per identity. This came at a real cost: ResNet18 has ~9.3x more parameters (11.2M vs 1.2M) and is ~9.3x larger on disk (42.64MB vs 4.61MB). Training time was longer for the custom CNN in this run (1077s vs 456s) because it needed more epochs (25 vs ~8) to reach its accuracy, even though each epoch is cheaper on the smaller model.

For a small, per-identity-data-constrained classification task like this one, the 10-point accuracy gap still favors the pretrained backbone despite its larger footprint. We carry `logs/resnet18_finetune_best.pt` forward as the classification component of the Milestone 2/3 detection pipeline. The custom CNN's recovery from 20% to 85% with tuned hyperparameters is itself a useful finding: it demonstrates that the from-scratch architecture is not inherently unable to learn this task, but is far more sensitive to augmentation strength and learning rate when trained on very little per-class data — exactly the practical trade-off that motivates using a pretrained backbone when data is scarce.